Chapter 1: Rational Choice and Preference Modeling

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Outline

1. Motivation and Basic Concepts
2. Preference Relations and Axioms
3. Utility Representation
4. Revealed Preference & Duality
5. Slutsky Equation & Welfare Measures
6. Exercises

Motivation: Why Study Rational Choice?

- **Core of microeconomic theory**: Understanding how agents make decisions under constraints.
- **Foundation for advanced topics**: Consumer theory, game theory, general equilibrium, etc.
- **Practical relevance**: Policy-making, business strategy, everyday decisions.

Key Goal

Build rigorous models to formalize **rational** decision-making and **preference relations**.

Key Questions

- How do individuals **rank** different outcomes?
- What properties must these rankings satisfy to be called *rational*?
- How can we derive a **utility function** from preferences (when possible)?
- How do we connect these ideas to **real-world** choice data and policy?

Preference Relation: Definition

Definition (Preference Relation)

Let X be a set of alternatives. A *preference relation* $\succeq$ on X is a binary relation such that for any $x, y \in X$,

$$x \succeq y \quad \text{means "}x\text{ is at least as good as }y\text{".}$$

- We write $x \succ y$ to denote *strict preference*: $x \succeq y$ but not $y \succeq x$.
- We write $x \sim y$ to denote *indifference*: $x \succeq y$ and $y \succeq x$.

Rational Preferences: Two Axioms

Definition (Completeness)

For all $x, y \in X$, either $x \succeq y$ or $y \succeq x$ (or both).

Definition (Transitivity)

For all $x, y, z \in X$, if $x \succeq y$ and $y \succeq z$, then $x \succeq z$.

Rational Preference

If a preference relation is **complete** and **transitive**, we call it *rational*.

Examples of Preferences

Example (Food Choice)

One might prefer pizza over salad or be indifferent if they value health and taste equally. This does not contradict rationality as long as the person can compare any two alternatives consistently.

Example (Leisure vs. Money)

A person might prefer \$10 over 1 hour of leisure, but also prefer 1 hour of leisure over \$5. This is still consistent with rationality, provided the comparisons are **transitive** *and* **complete**.

Beyond Completeness and Transitivity

Besides completeness and transitivity, we often add:

Definition (Continuity (Intuitive))

No abrupt "jumps" in preference. If $x \succ y$, then a slight change in x won't suddenly make $y \succ x$. Formally, it ensures the preference can be represented by a **continuous** utility function (see next slides).

Definition (Monotonicity)

More of a *good* is (weakly) better. If $x \geq y$ in every component, then $x \succeq y$.

Definition (Convexity)

If $x \sim y$, then $\lambda x + (1 - \lambda)y \succeq x$ for $0 < \lambda < 1$. Interpreted as a preference for "averages" over extremes.

Rational Choice

Definition

Given a **feasible set** $B \subseteq X$ (the set of possible alternatives under constraints), a rational decision-maker chooses $x \in B$ such that

$$x \succeq y \quad \text{for all } y \in B.$$

- In words: The agent chooses the **most preferred** option(s) from all feasible ones.
- Example: In consumer theory, B is determined by the **budget constraint**.

Example: Consumer Choice

- $X = \{(q_1, q_2) : q_1, q_2 \geq 0\}$ (two goods).
- Budget set: $B = \{(q_1, q_2) : p_1 q_1 + p_2 q_2 \leq m\}$.
- A *rational* consumer picks (q_1^*, q_2^*) that maximizes $\succeq$ within B.

Example (Concrete Setup)

If $p_1 = \$2, p_2 = \$1, m = \$10$, then $B = \{(q_1, q_2) : 2q_1 + q_2 \leq 10\}$. The consumer's choice depends on her preference ordering over (q_1, q_2).

Are Real Preferences Always Rational?

- **Incompleteness**: Some people cannot compare or rank certain alternatives.
- **Intransitivity**: Cyclical choices ($A \succ B, B \succ C, C \succ A$).
- **Context/Framing Effects**: Changing how a choice is presented might alter the preference.

Behavioral Economics

Human behavior can deviate from rational axioms, but classical microeconomics uses them as a *baseline* model.

Non-Continuous Preference: Dictionary Order

- A classical example of **discontinuous** (or "ill-behaved") preference is the *dictionary order* on $[0,1]^2$:

$$(x_1, x_2) \succ (y_1, y_2) \quad \text{if} \quad x_1 > y_1 \quad \text{or } (x_1 = y_1 \text{ and } x_2 > y_2).$$

- Intuitively, x_1 coordinate is "infinitely more important" than x_2 coordinate.
- This preference fails typical continuity assumptions; no single real-valued $u(\cdot)$ can represent it *continuously*.

Debreu's Theorem (1954)

Theorem (Representation Theorem)

If $\succeq$ is **rational** *and* **continuous** *on a suitable subset $X \subseteq \mathbb{R}^n$, then there exists a continuous utility function $u : X \to \mathbb{R}$ representing $\succeq$, i.e.*

$$x \succeq y \quad \Longleftrightarrow \quad u(x) \geq u(y),$$

for all $x, y \in X$.

- Strictly increasing transformations of u yield the *same* preference ordering.
- If continuity is absent (e.g., dictionary order), such a continuous $u(\cdot)$ generally does not exist.

Sketch of the Construction

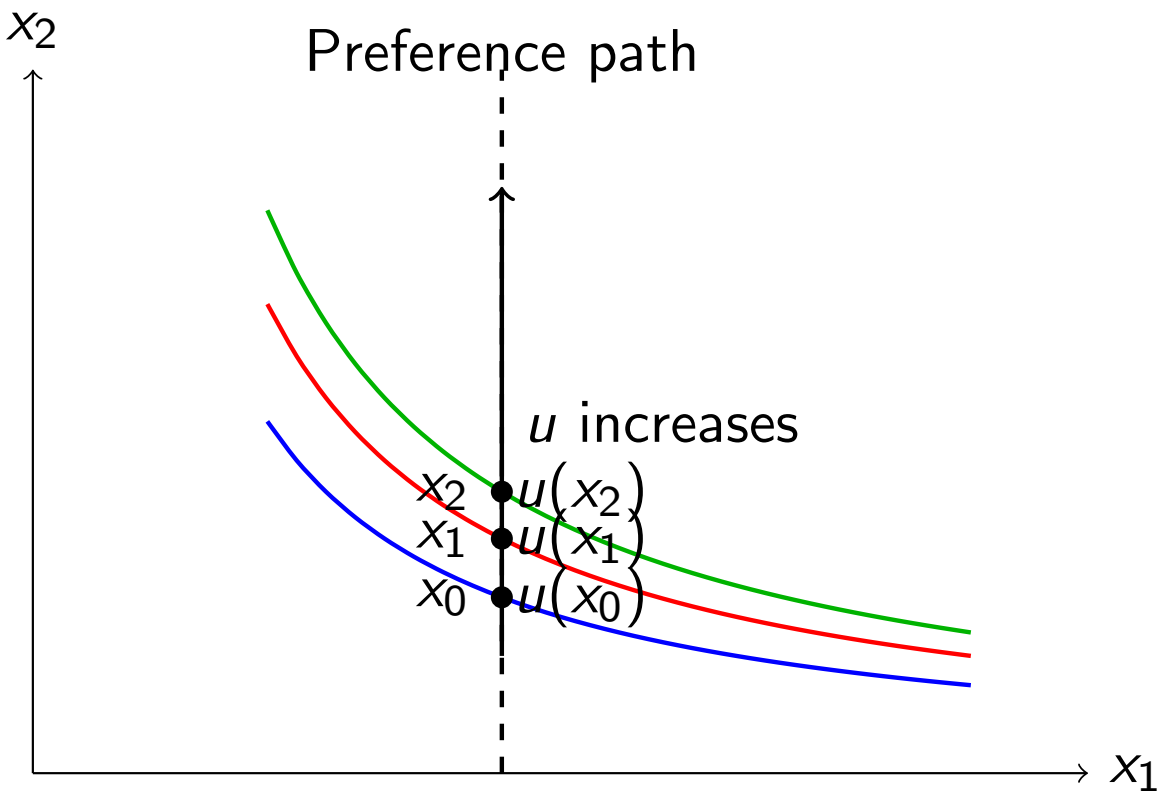
Intuitive idea:

1. **Fix** a reference alternative x_0.
2. For any x, find "how far" x is from x_0 along indifference curves or a preference path.
3. By continuity, we can assign a real number to each indifference curve in a *smooth* way.

Conclusion

This process *labels* each alternative with a real value $u(x)$ that **respects** the preference order.

Sketch of the Construction (Figure)



Ordinal vs. Cardinal Utility

- **Ordinal Utility**: Only the preference ranking matters. $u(x)$ can be replaced by any strictly increasing transformation $f(u)$.
- **Cardinal Utility**: Differences or ratios of utility levels have direct meaning (e.g., in expected utility theory).

Example (Example)

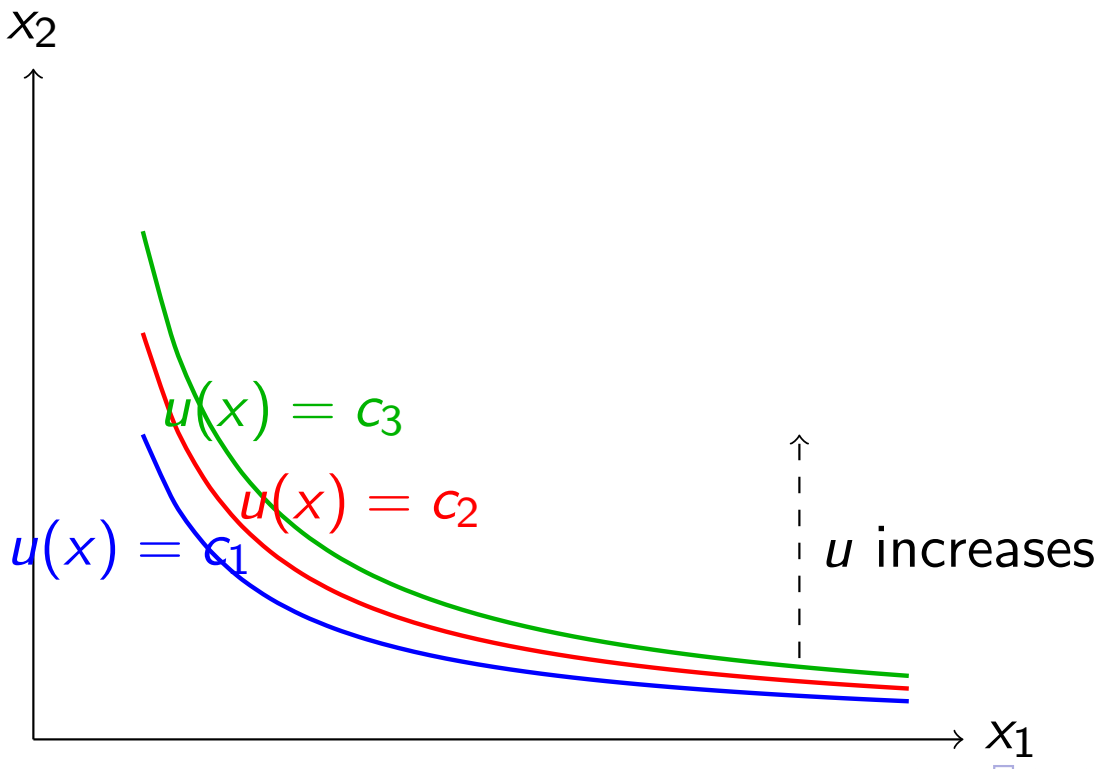
$u(x) = 2v(x) + 10$ *and* $v(x)$ *represent the same ordinal preference but different cardinal scales.*

Graphical Representation: Indifference Curves

- For $X \subset \mathbb{R}^2$, we often visualize preferences using **indifference curves**:

$$u(x_1, x_2) = \text{constant}.$$

- Each curve represents all bundles equally preferred.
- Higher curves correspond to *strictly better* bundles, under monotonicity.



Non-Convex Preferences: Runner's Example

- **Convex** preferences prefer (or at least do not mind) "averages" over "extremes."
- **Non-convex** preferences: Some runners only like integer distances (e.g., exactly 5 km or 10 km). A midpoint 7.5 km is *strictly worse* in their view.
- Graphically, the preferred sets can have "disconnected lumps," violating convexity.

Algebraic Example: Utility Maximization

Let $u(x_1, x_2) = x_1^\alpha x_2^\beta$ with $\alpha, \beta > 0$, subject to $p_1 x_1 + p_2 x_2 \leq m$.

- Lagrangian method or direct substitution yields:

$$x_1^* = \frac{\alpha}{\alpha + \beta} \frac{m}{p_1}, \quad x_2^* = \frac{\beta}{\alpha + \beta} \frac{m}{p_2}.$$

- If $\alpha, \beta > 0$, the preference is *strictly convex* and monotonic. No need to require $\alpha, \beta < 1$.

Marshallian (Walrasian) Demand

Utility Maximization Problem (UMP)

$$\max_{x \in \mathbb{R}^n_+} \quad u(x) \quad \text{subject to } p \cdot x \leq m.$$

- The solution $x(p, m)$ is called the **Marshallian** (or **Walrasian**) demand.
- It directly links (p, m) to the chosen bundle under *utility maximization*.

Revealed Preference Axioms

Revealed Preference uses *observed* choices to infer underlying preferences.

Definition (Weak Axiom of Revealed Preference (WARP))

If x is chosen over y when both are feasible, then in any other scenario where y is chosen (and x remains feasible), we get a contradiction of WARP.

$$\text{“}x \text{ revealed preferred to } y\text{”} \;\Rightarrow\; \text{not “}y \text{ revealed preferred to } x\text{”}.$$

- **Strong Axiom (SARP)** extends WARP to rule out cyclical revealed preferences.
- These axioms ensure *observed* demand can come from a rational preference relation.

Utility Maximization vs. Expenditure Minimization

Two **dual** problems:

(A) Utility Maximization Problem (UMP)

$$\max_{x \in \mathbb{R}^n_+} \quad u(x) \quad \text{subject to } p \cdot x \leq m.$$

(B) Expenditure Minimization Problem (EMP)

$$\min_{x \in \mathbb{R}^n_+} \quad p \cdot x \quad \text{subject to } u(x) \geq \bar{u}.$$

- (A) yields **Marshallian demand** $x(p, m)$.
- (B) yields **Hicksian (compensated) demand** $h(p, \bar{u})$.

Demand Functions

- **Marshallian demand**:

$$x(p, m) = \arg\max_x \{u(x) \ : \ p \cdot x \leq m\}.$$

- **Hicksian demand**:

$$h(p, \bar{u}) = \arg\min_x \{p \cdot x \ : \ u(x) \geq \bar{u}\}.$$

- They are related but serve different perspectives: one for *choice under a budget*, one for *cost of achieving a certain utility*.

Duality Theory

- **Indirect Utility**: $v(p, m) = \max_x \{u(x) : p \cdot x \leq m\}$.
- **Expenditure Function**: $e(p, \bar{u}) = \min_x \{p \cdot x : u(x) \geq \bar{u}\}$.
- **Key Relationship**:

$$v\big(p,\, e(p, \bar{u})\big) = \bar{u}, \quad e\big(p,\, v(p, m)\big) = m.$$

- This interplay is crucial for analyzing welfare changes (Slutsky, CV, EV, etc.).

Slutsky Equation: Introduction (1/4)

Recall: We have two types of demand functions:

1. $\mathbf{x}(p, m)$: the **Marshallian (Walrasian)** demand, solving

$$\max_x u(x) \quad \text{subject to } p \cdot x \leq m.$$

2. $\mathbf{h}(p, u)$: the **Hicksian (compensated)** demand, solving

$$\min_x p \cdot x \quad \text{subject to } u(x) \geq \bar{u}.$$

Key Question

When a price changes (say p_j), how do we decompose the total effect on $\mathbf{x}$ into a *substitution effect* and an *income effect*?

- $\mathbf{h}(p, u)$ holds **utility** constant (compensated demand).
- $\mathbf{x}(p, m)$ holds **income** (or budget) constant.

Slutsky Equation: Step-by-Step (2/4)

Idea: Suppose the price of good j changes from p_j to $p_j + dp_j$. We want to see how the demand for good i, x_i, changes.

1. **Substitution Effect**: Imagine we *compensate* the consumer so that they can stay at the same utility u^*. We look at $\mathbf{h}(p, u^*)$ and see how $\frac{\partial h_i}{\partial p_j}$ behaves.
2. **Income Effect**: After the price change, if we *do not* compensate, the consumer's real purchasing power shifts. This additional effect is captured by $-x_j \frac{\partial x_i}{\partial m}$.

$$\underbrace{\frac{\partial x_i}{\partial p_j}}_{\text{total change in Marshallian demand}} = \underbrace{\frac{\partial h_i(p, u^*)}{\partial p_j}}_{\text{substitution (Hicksian) effect}} - \underbrace{x_j \frac{\partial x_i}{\partial m}}_{\text{income effect}} .$$

Slutsky Equation: Step-by-Step (3/4)

$$\underbrace{\frac{\partial x_i}{\partial p_j}}_{\text{total change in Marshallian demand}} = \underbrace{\frac{\partial h_i(p, u^*)}{\partial p_j}}_{\text{substitution (Hicksian) effect}} - \underbrace{x_j \frac{\partial x_i}{\partial m}}_{\text{income effect}} .$$

Interpretation

- The first term isolates how the consumer *substitutes* between goods when compensated to stay at the same utility.
- The second term shows how reduced/increased *purchasing power* further alters demand.

Slutsky Equation: Why It Matters (4/4)

- **Policy Implications**: If the government changes a tax (thus affecting a price), the Slutsky equation helps us see how much of the demand change is due to pure substitution vs. income effects.
- **Sign of Substitution Effect**:
  - Usually negative for normal goods (if p_j goes up, we substitute away).
  - But the **income effect** can be positive or negative depending on whether the good is normal or inferior.
- **Building Block** for advanced results:
  - Deriving **compensated demand curves**, $\mathbf{h}(p, u)$.
  - Analyzing **cost-of-living** indices and welfare measures (CV, EV).

Takeaway

Slutsky's decomposition is the backbone of consumer theory, clarifying the structure of demand responses to price changes.

Introducing CV and EV

Motivation: When a price changes (e.g., from $\mathbf{p^0}$ to $\mathbf{p^1}$), we want to measure:

- **How much better or worse off** is the consumer?
- **How much money** is this price change worth?

Key Differences:

- Both CV and EV measure welfare changes, but from **different reference points** (new vs. old prices).
- CV is often used in **compensation arguments** (post-change scenario).
- EV can reflect **willingness to pay** for avoiding the change.

Conclusion

Both CV and EV refine our welfare analysis by showing how price changes translate into *monetary* gains or losses from the consumer's perspective.

Welfare Measures: Compensating Variation (CV)

Definition: The **Compensating Variation (CV)** is defined as the additional income that a consumer would need at the **new** prices to reach their **original** utility level. In formula,

$$\text{CV} = e(p^1, u^0) - e(p^0, u^0),$$

where:

- $e(p, u)$ is the **expenditure function**—the minimum income required to achieve utility u at prices p.
- p^0 are the original prices, and p^1 are the new prices.
- u^0 is the initial (original) utility level (with $e(p^0, u^0) = m$, the original income).

Interpretation: CV measures how much extra money a consumer would need at the new price vector p^1 to be as well off as before the price change.

Welfare Measures: Equivalent Variation (EV)

Definition: The **Equivalent Variation (EV)** is defined as the amount of income that, at the **original** prices, would have to be taken away from the consumer to reduce their utility to the level achieved under the new prices. That is,

$$\text{EV} = e(p^0, u^0) - e(p^0, u^1),$$

where:

- u^0 is the original utility level.
- u^1 is the new utility level after the price change.

Interpretation: EV reflects the consumer's willingness to pay (or the monetary equivalent) to avoid the price change—measured in terms of income adjustments at the original prices.

Summary of Key Concepts

- **Rational preference**: complete + transitive.
- **Continuity** ensures existence of a continuous utility representation.
- **Ordinal** vs. **cardinal** utility.
- **Marshallian (Walrasian) demand** vs. **Hicksian (compensated) demand**.
- **Slutsky equation**, **CV**, **EV** for analyzing welfare impacts of price changes.

Exercises (1)

1. **Preference Axioms**: Prove that if $\succeq$ is transitive and complete, then $\succ$ and $\sim$ inherit transitivity (where applicable).
2. **Utility Construction**: Construct a utility function for $\succeq$ defined by $x \succeq y$ iff $\sum x_i \geq \sum y_i$ (for $x, y \in \mathbb{R}^n_+$). Is it continuous?
3. **Dictionary Order**: Show briefly why dictionary order on $[0,1]^2$ lacks a continuous utility representation.

Exercises (2)

1. **Revealed Preference**: State WARP. Illustrate with a numerical example how violating WARP implies inconsistency.
2. **Hicksian Demand**: Solve an expenditure minimization problem for a Cobb-Douglas $u(x_1, x_2) = x_1^{\alpha} x_2^{\beta}$ and derive $h(p, \bar{u})$.
3. **Duality**: Show $e\big(p,\ v(p, m)\big) = m$ using definitions of $e(\cdot)$ and $v(\cdot)$.

Exercises (3)

1. **Slutsky Decomposition**: Derive the Slutsky equation for a 2-good case. Interpret each term.
2. **CV & EV**: For a simple price change scenario (e.g., p_1 from 2 to 3, p_2 fixed, m fixed), compute CV and EV using the expenditure function approach.
3. **Duality for a Cobb-Douglas Utility Function**: Consider a consumer with the Cobb-Douglas utility function

$$u(x_1, x_2) = x_1^{\alpha} x_2^{\beta}, \quad \alpha, \beta > 0.$$

Answer the following:

Exercises (3)

(a) **Expenditure Function:** Derive the expenditure function

$$e(p_1, p_2, \bar{u})$$

that represents the minimum income required to achieve a utility level $\bar{u}$.

(b) **Indirect Utility Function:** Derive the indirect utility function

$$v(p_1, p_2, m)$$

corresponding to the consumer's utility maximization problem.

(c) **Duality Verification:** Verify the duality relationship by showing that

$$e\Big(p_1, p_2, v(p_1, p_2, m)\Big) = m.$$

References

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Extended Topics to Explore

- Slutsky Equation in multi-good context and applications.
- Risk and Uncertainty: Expected Utility, Risk Aversion.
- Intertemporal Choice, Behavioral Time Preference.
- General Equilibrium Theory and Welfare Theorems.

Coming Up Next...

We will delve deeper into consumer demand, production theory, and possibly general equilibrium.

Concluding Remarks

- Rational choice + continuity underpin the standard utility-based consumer model.
- Slutsky analysis and CV/EV are powerful tools to assess **welfare** effects of price/income changes.
- Real-world violations exist, but the classical model remains our core reference framework.

Thank you for your attention!

Q&A